

$$I(2,2) = -2$$

Age	Yes P	No N	$I(P, N)$
≤ 30	2	3	0.971
31-40	4	0	0
> 40	3	2	0.971
Total	9	5	0.940

$$\begin{aligned}
 I(2,3) &= -\sum_{i=1}^m P_i \log_2(P_i) \\
 &= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \\
 &= -\frac{2}{5} (-1.322) - \frac{3}{5} (-0.737) \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 I(4,0) &= -\sum_{i=1}^m P_i \log_2(P_i) \\
 &= -\frac{4}{4} \log_2\left(\frac{4}{4}\right) - \frac{0}{4} \log_2\left(\frac{0}{4}\right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 I(3,2) &= -\sum_{i=1}^m P_i (\log_2(P_i)) \\
 &= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \\
 &= -\frac{3}{5} (-0.737) - \frac{2}{5} (-1.322) \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 I(9,5) &= \frac{9}{41} \log_2\left(\frac{9}{41}\right) - \frac{5}{41} \log_2\left(\frac{5}{41}\right) \\
 &= \frac{9}{41} (-0.6374) - \frac{5}{41} (-1.4854) \\
 &= 0.940
 \end{aligned}$$

Index

$$\begin{aligned} I(2,2) &= -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \\ &= -\frac{2}{4} (-1) - \frac{2}{4} (-1) = \frac{2}{4} + \frac{2}{4} = \frac{4}{4} = 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} I(9,5) &= -9/14 \log_2 (9/14) - 5/14 \log_2 (5/14) \\ &= -9/14 \log_2 (-0.634) - 5/14 \log_2 (-1.4854) \\ &= 0.192 \end{aligned}$$

$$\text{Gain (income)} = 0.029$$

$$\text{Gain (student)} = 0.151$$

$$\text{Gain (credit-rating)} = 0.048$$

→ Gain ratio :- = Gain A / Split Info

$$\text{Split Info} = - \sum_{j=1}^V \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$\begin{aligned} \text{Split Info age} &= -\frac{5}{14} \log_2 \left(\frac{5}{14} \right) - \frac{4}{14} \log_2 \left(\frac{4}{14} \right) - \left(\frac{5}{14} \right) \log_2 \left(\frac{5}{14} \right) \\ &= -5/14 \log_2 (-1.4854) - 4/14 \log_2 (-1.8074) \\ &\quad - (5/14 \log_2 (-1.4854)) \\ &= 1.58 \end{aligned}$$

$$\text{Gain Ratio} = \frac{\text{Gain Split}}{\text{Split Info}} = \frac{0.246}{1.58} = 0.1556$$

$$\begin{aligned} \Rightarrow \text{Split info (income)} &= - \sum \frac{|D_j|}{|D|} \log_2 \frac{|D_j|}{|D|} \\ &= \frac{4}{14} \log_2 \frac{4}{14} + \frac{6}{14} \log_2 \frac{6}{14} + \frac{4}{14} \log_2 \frac{4}{14} \\ &= -\frac{4}{14} (-1.8074) - \frac{6}{14} (-1.2224) - \frac{4}{14} (-1.8074) \\ &= 1.557 \end{aligned}$$

$$GR = \frac{\text{Gain}}{\text{Split info}} = \frac{0.029}{1.557} = 0.019$$

$$(ii) \text{ Info } A(D) = \sum_{i=1}^n \frac{|D_i|}{|D|} \times \text{info}(D_i)$$

$$\begin{aligned} \text{Info age}(D) &= 5/14 \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \\ &= 0.346 + 0 + 0.346 \\ &= 0.692 \end{aligned}$$

$$\begin{aligned} (iii) \text{ Gain} &\Rightarrow \text{Info}(D) - \text{Info age}(D) \\ &= 0.940 - 0.692 \\ &= 0.248 \end{aligned}$$

Income	Yes P	No N	Info(Pn)
Low	3	1	0.811
Medium	4	2	0.917
High	2	2	1
Total	9	5	

$$\begin{aligned} I(3,1) &= -\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \\ &= -\frac{3}{4}(-0.41) - \frac{1}{4}(-2) \\ &= 0.811 \end{aligned}$$

$$\begin{aligned} I(4,2) &= -\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \\ &= -\frac{4}{6}(-0.585) - \frac{2}{6}(-1.585) \\ &= 0.917 \end{aligned}$$

Measure

Gain Index

$$\text{Gini}(D) = 1 - \sum_{j=1}^n p_j^2$$

$$\text{gini}_A(D) = \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2)$$

Finding Gini Index Based on.

$$\begin{aligned} \text{gini} &= 1 - \left[9/14 + 5/14 \right]^2 \\ &= 1 - (0.642 + 0.1257) = [1 - 0.7677] \\ &= 0.2323 \approx 0.23 \end{aligned}$$

Partition the data set w.r. to Income.

$$D_1 = \{\text{Low, Medium}\}$$

$$D_2 = \{\text{High, Medium}\}$$

$$\begin{aligned} \text{Gini}_{\text{income}}(D) &= \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2) \\ &= \frac{10}{14} \left[1 - \left(\frac{7}{10} \right)^2 - \left(\frac{3}{10} \right)^2 \right] + \frac{4}{14} \left[1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right] \\ &= \frac{10}{14} \left[1 - \frac{49}{100} - \frac{9}{100} \right] + \frac{4}{14} \left[1 - \frac{4}{16} - \frac{4}{16} \right] \\ &= \frac{10}{14} \left[\frac{100 - 49 - 9}{100} \right] + \frac{4}{14} \left[\frac{16 - 4 - 4}{16} \right] \\ &= \frac{10}{14} \left[\frac{42}{100} \right] + \frac{4}{14} \left[\frac{8}{16} \right] \\ &= \frac{1}{14} \left[\frac{42 \cdot 10}{10} \right] + \frac{2}{14} \\ &= \frac{1}{14} \left[\frac{420}{10} \right] + \frac{2}{14} = \frac{1}{14} \left[\frac{420}{10} + 2 \right] \\ &= \frac{1}{14} \left[\frac{422}{10} \right] = \frac{422}{140} \approx 3.014 \end{aligned}$$

$$Gini' = (L/m) =$$

$$(LH) \rightarrow M$$

$$gini'(Income) = \frac{|D_1|}{|D|} gini'(D_1) + \frac{|D_2|}{|D|} gini'(D_2)$$

$$gini'(L/m) = 0.458$$

$$gini'(M/h) = 0.450$$

$H = 0.443 \checkmark \rightarrow$ High we can Buy the Computer

$$M = 0.458$$

$$L = 0.450$$

{Low, Medium}

{High}

{0.443}

{Low, High}

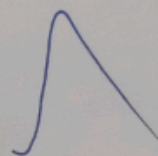
{Medium}

{0.458}

{High, Medium}

{Low}

{0.450}



4/

Sol:-

Actual	Precision	
	TP	FN
	FP	TN

- 2) Assume there are 100 images, 30 of them depict a cat the rest do not. A Machine Learning Model Predicts the occurrence of a cat in 25 of 30 cat images. it also Predicts absence of a cat in 50 of the 70 no cat image. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score.

Given

Total images = 100

Cat images = 30

No. Cat image = 70

→ Model Predicts correctly 25 of 30 cat images

→ Model Predicts No Cat Correctly for 50 of 70 no-Cat images

Actual/ Predicted	Cat	No Cat
Cat	TP = 25	FN = 5
No Cat	FP = 20	TN = 50

$$TN = 50$$

$$FP = 70 - 50 = 20$$

$$\rightarrow \text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 20} = \frac{25}{45} = 0.556$$

$$= 0.556 \times 100$$

$$= 55.56\%$$

$$\rightarrow \text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{25}{25 + 5} = \frac{25}{30} = 0.833$$

$$= 0.833 \times 100$$

$$= 83.33\%$$

$$\rightarrow F_1 \text{ score} = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2(0.556)(0.833)}{0.556 + 0.833}$$

$$= 0.6667$$

$$F_1 = 66.7\%$$

- 3) An email service Provider wants to classify email emails as spam (or) Not spam using features
- Contains "offer" (Yes/No)
 - Contains "free" (Yes/No)
 - Email length (Short/Long)

sol: Given Dataset:

offer	free	length	class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Total emails = 6

Spam = 3

Not Spam = 3

Confusion Matrix:-

1. Prior Probabilities:-

Probability of spam = $\frac{\text{Number of spam emails}}{\text{total emails}}$

$$P(\text{spam}) = \frac{3}{6} = \frac{1}{2}$$

Probability of not spam

$$P(\text{not spam}) = \frac{3}{6} = \frac{1}{2}$$

2. Conditional Probabilities:-

A. For spam

There are 3 spam emails

Offer

$$P(\text{offer} = \text{yes} | \text{spam}) = \frac{2}{3}$$

$$P(\text{offer} = \text{no} | \text{spam}) = \frac{1}{3}$$

Free

$$P(\text{free} = \text{yes} | \text{spam}) = \frac{3}{3} = 1$$

$$P(\text{free} = \text{no} | \text{spam}) = \frac{0}{3} = 0$$

length

$$P(\text{short} | \text{spam}) = \frac{2}{3}$$

$$P(\text{long} | \text{spam}) = \frac{1}{3}$$

3. Classify the New Email

Given

offer = Yes, Free = No, length = short

$$P(\text{spam} | x) \propto P(\text{spam}) P(\text{offer} = \text{yes} | \text{spam})$$

$$= \frac{1}{2} \times \frac{2}{3} \times \frac{0}{3} \times \frac{2}{3} = 0$$

Confusion Matrix:-

Actual class \ Predicted class	Cancer yes	Cancer = no	Total
Cancer = yes	90 TP	210	300 → P
Cancer = no	140 FN	9560 TN	9700 → N
Total	230	9770	10000

$$\begin{aligned}
 1. \text{ Accuracy} &= \frac{TP + TN}{P} \\
 &= \frac{90 + 9560}{10000} \\
 &= \frac{90.956}{100} \times 100 \\
 &= 90.956 \quad 96.5
 \end{aligned}$$

$$\begin{aligned}
 4) \text{ Specificity} &= \frac{TN}{N} \\
 &= \frac{9560}{9700} \\
 &= 0.0144 \times 100 \\
 &= 0.98 \times 100 \\
 &= \underline{98.5}
 \end{aligned}$$

$$\begin{aligned}
 2. \text{ Error Rate} &= 1 - \text{Accuracy} \\
 &= 1 - 96.5 \\
 &= 1 - 0.965 \\
 &= 0.035 \times 100 \\
 &= 3.5
 \end{aligned}$$

$$5) \text{ Precision} = \frac{TP}{TP + FP}$$

$$\begin{aligned}
 &= \frac{90}{90 + 140} \\
 &= \frac{90}{230}
 \end{aligned}$$

$$\begin{aligned}
 3) \text{ Sensitivity} &= \frac{TP}{P} \\
 &= \frac{90}{300} \\
 &= 0.3 \times 100 \\
 &= 30\%
 \end{aligned}$$

$$\begin{aligned}
 &= 0.39 \times 100 \\
 &= \underline{39.13}
 \end{aligned}$$

Euclidean Manhattan and
ski distance.

two points

formula, Manhattan

Not spam score

$$P(\text{Not spam} | x) \propto P(\text{Not spam}) \cdot P(\text{offer} = y)$$

$$= \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3} = \frac{1}{18} = 0.0556$$

— 0 —

Heard
u/s/x

CO3 - ATL-2

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Code

→ Entropy :-

$$H(Y) = - \sum_{i=1}^m P_i \log(P_i)$$

→ Information :-

$$\text{info}(D) = - \sum_{i=1}^m P_i \log_2(P_i)$$

$$\text{Info A}(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{info}(D_j)$$

→ Gain :-

$$\text{Info}(D) - \text{Info A}(D)$$

→ Assume

class P :- buys - Computer = "Yes"

class N :- buys computer = "No"